Recitaion 08

List and Tuples

Exercise 1 | Predict the output

1	`T = (1, 2, 3)` `T[0] = 3`	
2	`L = [1, 2, 3]` `print(L.append([4, 5]))`	

3	`L = [4, 2, 1, 3]`
	`L.sort()`
	`print(L)`	

4	`L = ['a', 'b', 'c']`
	`L.extend(['d', 'e'])`
	`L.extend('fg')`
	`print(L)`	

5	`L = ["apple", "cat", "banana", "dog"]`
	`L1= [len(x) for x in L]`
	`L2= [x for x in L if len(x) > 4]`
	`print(L1)`
	`print(L2)`	

6	`t = [1, 3, 5, 7, 9]`
	`for i in t:`
	`  if i%2:`
	`  t.remove(i)`
	`print(t)`	

7	`def myappend(L):`
	`  L.append(3)`
	`  return L`
	`x = [1, 2]`
	`y = myappend(x) # x = [1,2,3]`
	`y.append(4) # y = [1,2,3,4]`
	`print(x) # x = [1,2,3,4]`	

Exercises 2 | Total Sales

Design a program that asks the user to enter a store's sales for each day of the week. The amounts should be stored in a list. Use a loop to calculate the total sales for the week and display the result.

Exercises 3 | Lottery Number Generator

Design a program that generates a seven-digit lottery number. The program should generate seven random numbers, each in the range of 0 through 9, and assign each number to a list element. Then write another loop that displays the contents of the list

Exercises 4 | Number Analysis Program

Design a program that generates at random a series of 20 numbers in the range of 1 through 100. The program should store the numbers in a list and then display the following data:

- The lowest number in the list
- The highest number in the list
- The total of the numbers in the list
- The average of the numbers in the list

Exercises 5 | Larger Than n

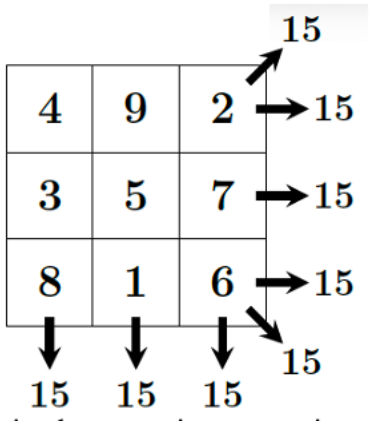
In a program, write a function that accepts two arguments: a list, and a number n. Assume that the list contains numbers. The function should display all of the numbers in the list that are greater than the number n.

Exercises 6 | Lo Shu Magic Square

The Lo Shu Magic Square is a grid with 3 rows and 3 columns. It has the following properties:

- The grid contains the numbers 1 through 9 exactly.
- The sum of each row, each column, and each diagonal all add up to the same number.

This is shown in the following Figure:



In a program you can simulate a magic square using a two-dimensional list. Write a function that accepts a two-dimensional list as an argument and determines whether the list is a Lo Shu Magic Square. Test the function in a program.

Exercise 7 | Grades

This question is about data related to students enrolled in a course and their grades during a semester.

The data about students and their grades is stored in a list of lists where each sublist represents one student. All sublists have the following seven (7) pieces of information:

- the student id as a string,
- the first name of the student as a string,
- the last name of the student as a string,
- the grades of four (4) tests as strings.

A sample list is shown below:

```
student_grades = [['2345243', 'John', 'Doe', '79', '89', '49', ’79'],
                  ['5634423', 'Bob', 'Smith', '90', '90', '80', '90'],
                  ['1433145', 'Jane', 'Roe', '61', '71', '81', '81'],
                  ['8754543', 'Joe', 'Public', '82', '93', '92', ’92'],
                  ['3142354', 'Jean', 'Martin', '95', '37', '39', '98']]
```

1) Write a function named `computeAverage()` that takes a parameter named `studentList` in the same format as list `student_grades` and returns a list of list where each sublist represents one student. All sublists consist of the following information:

- the student id as a string,
- the first name of the student as a string,
- the last name of the student as a string,
- the average grade for all four (4) tests as a string.

Example output:

Call	`computeAverage(student_grades)`

Returned value	`[['2345243', 'John', 'Doe', '74.0'],`
	` ['5634423', 'Bob', 'Smith', '87.5'],`
	` ['1433145', 'Jane', 'Roe', '73.5'],`
	` ['8754543', 'Joe', 'Public', '89.75'],`
	` ['3142354', 'Jean', 'Martin', '67.25']]`

2) Write a function `highestAverage()` that takes a parameter named `studentList` in the same format as list `student_grades` and returns a list with the information of the best performing student.

Example output:

Call	`highestAverage(student_grades)`
Returned value	`['8754543', 'Joe', 'Public']`

Exercise 8 | Random Attendance System

Write a program that stores student records in a list. Each record is a tuple containing a student's name and absence count, for example:

```
Stu_list = [("Alice", 0), ("Bob", 0), ("Charlie", 0)]
```

Write the following functions:

1. `register_students()`
   Ask the user to enter all student names. Set each student's absence count to 0 and return the list of records.

2. `take_attendance(Stu_list, n)`
   Randomly select `n` different students. For each selected student, display their name and ask whether they are present. If absent, increase their absence count by one. Update the original list. Assume attendance input is `"P"` for present or `"A"` for absent.

3. `check_absences(student)`
   Ask for a student's name and display their absence count. If the name is not found, display `"Student not enrolled."`.

4. `main()`
   Register the students, then repeatedly display this menu:
   1. Take attendance
   2. Check a student's absence count
   3. Quit

```
sample output:
Enroll students……

1. Take attendance
2. Check a student's absence count
3. Quit
Choose an option: 1
How many students should be selected? 2
Is Bob present? (P/A): A
Is Alice present? (P/A): P

1. Take attendance
2. Check a student's absence count
3. Quit
Choose an option: 2
Enter the student's name: Bob
Bob: 1 absence(s)

1. Take attendance
2. Check a student's absence count
```

```
3. Quit
Choose an option: 1
How many students should be selected? 2
Is Charlie present? (P/A): P
Is Bob present? (P/A): A

1. Take attendance
2. Check a student's absence count
3. Quit
Choose an option: 2
Enter the student's name: Bob
Bob: 2 absence(s)

1. Take attendance
2. Check a student's absence count
3. Quit
Choose an option: 2
Enter the student's name: David
Student not enrolled.

1. Take attendance
2. Check a student's absence count
3. Quit
Choose an option: 3
```